

Stroke Data Prediction Analysis Using Machine Learning Techniques

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Abstract—According to the updated definition of stroke from the American Heart Association, a stroke is an episode of acute neurological dysfunction presumed to be caused by ischemia or hemorrhage, persisting ≥ 24 hours or until death. It ranked third as one of the leading causes of death globally in 2021. This paper will cover the correlation between attributes such as gender, age, hypertension, heart disease, marital status, work type, residence type, etc to the likelihood of getting a stroke. The data will be processed using statistical models and machine learning models. The statistical model used in this study would be scatter plots with linear regression and chi square test. The machine learning model would be decision trees and random forests. And for further classification neural networks are used. According to this study the result shows that there is a correlation between the attributes and the likelihood of developing stroke. The results of this study show that Neural Network yielded the highest prediction accuracy and robustness among the models tested. Key contributing factors to stroke prediction include age, hypertension, heart disease, average glucose level, BMI, and smoking status. This indicates that machine learning-based classification can serve as a valuable tool in early stroke risk detection.

Keywords—Stroke, Classification, Data Preprocessing, Statistical Model, Linear Regression, Chi Square Test, Machine Learning, Decision Tree, Random Forest, Neural Network

I. INTRODUCTION

A. Background

Stroke is one of the leading causes of death and long-term disability worldwide. The disease occurs when the blood supply to the brain is interrupted or reduced, leaving brain tissue deprived of oxygen and nutrients. Stroke is often directly related to unhealthy lifestyle habits. Lifestyles such as smoking, consumption of foods high in fat and salt, lack of physical activity, and excessive stress can increase the risk of high blood pressure, high

cholesterol, and diabetes-the main factors that cause stroke. When blood flow to the brain is disrupted due to blockage or rupture of blood vessels, brain cells begin to die within a short period of time. If left untreated, this condition can cause permanent damage to the brain and even death. Therefore, maintaining a healthy lifestyle is essential in preventing stroke and its fatal effects. Early detection and prevention are crucial in reducing the risk of stroke, especially by utilizing appropriate technology and data analysis [1].

The development of information technology has brought significant progress in the field of health, one of which is the application of data science and machine learning for predictive analysis. Through the utilization of publicly available health datasets, such as the stroke prediction dataset, researchers and practitioners can develop predictive models to assist the diagnosis and medical decision-making process [2].

This study aims to analyze individual health data and identify factors that have an association with stroke risk using statistical and classification approaches. Thus, this study can contribute to helping medical personnel and the general public understand important factors in stroke prevention [3], [4], [5].

B. Research Objective

The purpose of this study is to describe the characteristics of patients based on the variables available in the stroke prediction dataset, as well as conduct statistical analysis to determine the relationship between risk factors such as age, hypertension, heart disease, smoking habits, and blood glucose levels to the chance of having a stroke. In addition, this study also aims to develop a basis for predictive modeling that can be used to identify individuals at high risk of stroke using a classification approach.

C. Problem Statement

The problem statement in this study is how the general characteristics of patients in the stroke prediction dataset, what factors contribute chances of someone having a stroke, and how statistical analysis and data visualization can be used to evaluate the relationship between variables in the context of stroke risk prediction.

D. Aim

This research is expected to provide benefits for academics as a reference in the development of health

data-based disease prediction studies, for the general public as a means of education to understand stroke risk factors, and for medical personnel as supporting information in making more objective and systematic data-based clinical decisions.

II. LITERATURE REVIEW

The development of information and computing technology in healthcare has led to the utilization of machine learning (ML) as an innovative approach in the early detection and prediction of diseases, including stroke, which is a leading cause of death and long-term disability worldwide. In this study, various machine learning-based classification methods were used to model the risk of stroke based on patients' clinical and lifestyle data.

Some popular algorithms used in stroke prediction studies include Decision Tree, Random Forest, and Neural Network. Each algorithm has advantages in terms of accuracy, interpretability, and efficiency in handling complex and diverse medical data.

Previous literature has extensively evaluated the effectiveness of these algorithms in the context of medical classification, both using structured datasets such as patient clinical data (age, blood pressure, and glucose levels), and unstructured data such as EEG (Electroencephalogram) brain signals. Based on the findings in various studies, ensemble methods such as Random Forest (RF) show excellent classification performance, followed by rule-based methods such as Decision Tree (DT), as well as complex predictive models such as Neural Network (NN), especially when applied to non-linear data or biological signals.

A. Decision Tree

Decision Tree is one of the most popular classification methods in Machine Learning (ML) due to its ability to generate models that are easy to understand and interpret. This model divides a dataset into subsets based on attribute values, and creates a tree-like structure consisting of decision nodes and leaf nodes.

Desy I. Puspitari, Windha M. P. Dhubita, Al-Fath R. Kholdani. (2021). found used to classify whether an individual has a risk of stroke based on variables such as age, hypertension, glucose levels, and smoking status. Decision Tree method successfully achieved an accuracy of 95.90% in stroke risk classification. In addition, Decision Tree also has the advantage of interpretability-each path from root to leaf represents an explicit classification rule, which makes it suitable for clinical applications where decision transparency is critical [6].

B. Random Forest

Random Forest (RF) is an ensemble method that combines a number of decision trees to improve prediction accuracy and stability. Each tree is trained on a different subset of data, and the final prediction is determined by majority voting.

Md. Khalilur Rahman, Md. Ashikur Rahman Khan, Ishtiaq Ahammad, Joysri Rani Das. (2025). showed that Random Forest achieved the highest accuracy of 97.85%, with equally high precision and recall, signifying its robustness in minimizing false negatives which is important in medical diagnosis. Random Forest is also known to have better tolerance to overfitting than Decision Tree, as it

reduces variance through aggregation of multiple tree models [7].

C. Neural Network

Neural Network (NN) is a model inspired by the workings of the human brain and consists of layers of connected artificial neurons. These models are capable of learning complex non-linear patterns in data, making them suitable for clinical predictions such as stroke.

Akhtam F. Sawani, Mohammed Awad, Radwan Qasrawi, Mohammad Sowon. (2023). The use of EEG as the primary signal for stroke diagnosis allows the application of neural network models, particularly with algorithms such as *Modified XGBoost* and *Multilayer Perceptron* (MLP). Although these models tend to be more complex and difficult to interpret, neural networks have the advantage of handling data with many features and non-linear interactions, such as EEG biological signals or multi-dimensional medical data.

Nevertheless, the main limitations of Neural Networks in clinical applications are transparency and long training time. Therefore, in this study, Neural Network was used as a comparison model and not as the main model, although its performance remains competitive. [8]

III. METHODOLOGY

A. Data Acquisition

The data used in this study comes from a public dataset retrieved on the *Kaggle* platform. The dataset contains individual health information, including demographic data, disease history, lifestyle, and medical status related to the chance of stroke. The data were collected as CSV files and processed using statistical analysis software and R and Python programming. This data did not require a direct data capture process as it was available in ready-to-analyze form.

B. Sample and Population

The population in this study includes all individuals whose data are included in the stroke dataset. The sample used was 5109 individuals after a cleaning process, which removed entries with invalid gender ("Other") and BMI values that were empty or of type "N/A".

C. Research Variables

In this study, a number of variables were used that reflect the demographic characteristics, health conditions, and lifestyle of individuals. These variables consist of numerical and categorical data that can be used to evaluate the relationship between various risk factors and stroke incidence. The details of each variable are described in the following table:

Number	Name	Type of Data
X1	Age	Numerical
X2	Hypertension	Binary (0 or 1)
X3	Heart Disease	Binary (0 or 1)
X4	Married	Binary (Yes or No)
X5	Work Type	Categorical (input type)

X6	Avg Glucose Level	Numerical
X7	BMI	Numerical
X8	Smoking Status	Categorical
Y	Stroke	Binary (0 or 1)

D. Analysis Methods

Data analysis was carried out in several stages as follows:

1. Pre-Processing Data

In the data preprocessing stage, the first step was to remove invalid values, such as entries with a gender of “Other” that were not included in the analysis category. Furthermore, data with unavailable BMI values or containing “N/A” were also removed to ensure consistency and accuracy of the analysis. In addition, the data type on each variable was changed to the appropriate form, such as converting categorical data to factor type and numerical data remained in number format for further statistical analysis.

2. Descriptive Analysis

Use frequency tables, percentages, means, medians, and data visualizations such as pie charts, bar charts, and boxplots for each variable.

3. Correlation Analysis and Statistical Test

In the inferential analysis, Pearson or Spearman correlation tests were used to measure the association between numerical variables, such as the association between age and body mass index (BMI). Meanwhile, to test the association between categorical variables, such as between gender and stroke incidence, the Chi-Square test was used. In addition, hypothesis tests such as the t-test or ANOVA were applied to determine whether there were significant differences between certain groups of data on the numerical variables observed.

4. Classification

This study was designed for use in binary classification (0 = no stroke, 1 = stroke) using model testing and training algorithms such as:

- Linear Regression
- Decision Tree
- Random Forest
- Neural Network

E. Tools and Software

Data analysis was conducted using a combination of:

- Rstudio - for visualization and exploratory statistical analysis
- Python (Scikit-learn, Pandas, Matplotlib) - for modelling classification
- Microsoft Excel - for the initial cleanup and simple exploration phase

IV RESULT AND ANALYSIS

In this section, we present and analyze the results we obtained from our implementation of Linear Regression, Decision Tree, Random Forest and Neural Network models for stroke data prediction. These metrics are employed to

evaluate how accurate and effective each model performs by model training and testing.

A. Categorical Variables (Pie-Chart)

1. Gender

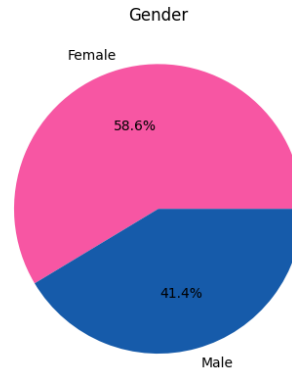


Fig. 1. Gender

2. Martial Status

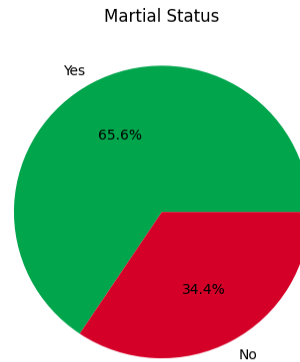


Fig. 2. Martial Status

3. Work Type

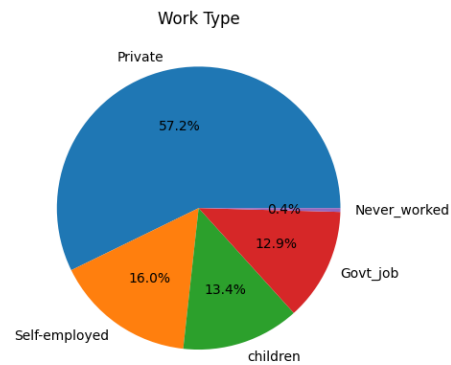


Fig. 3. Work Type

4. Residence Type

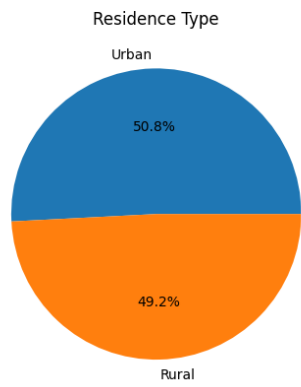


Fig. 4. Residence Type

5. Smoking Status

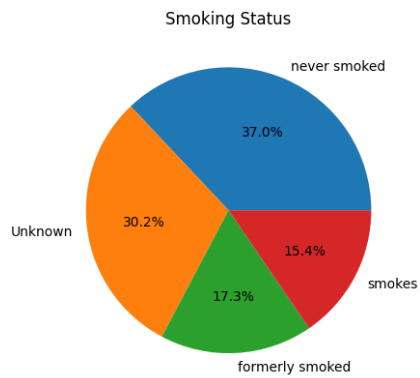


Fig. 5. Smoking Status

6. Hypertension

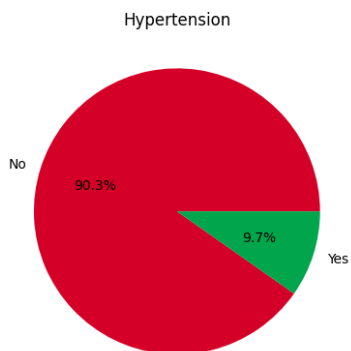


Fig. 6. Hypertension

7. Heart Disease

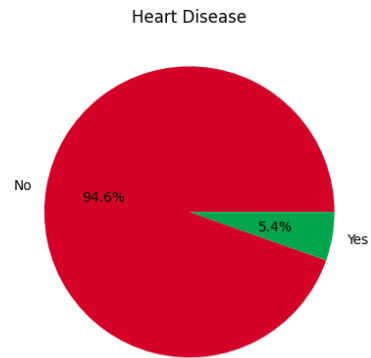


Fig. 7. Heart Disease

8. Stroke

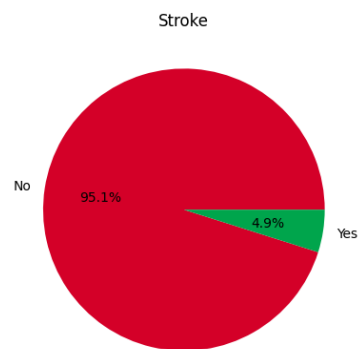


Fig. 8. Stroke

B. Numerical Variables (Box-Plot)

1. Age

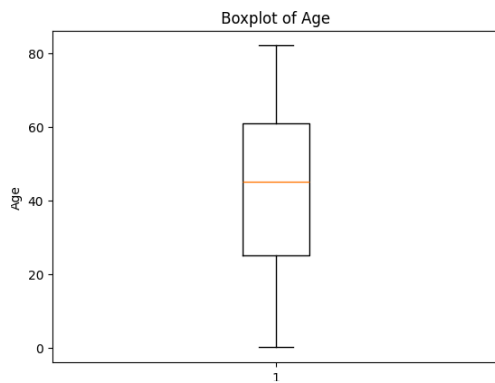


Fig. 1. Age

2. Average Glucose Level

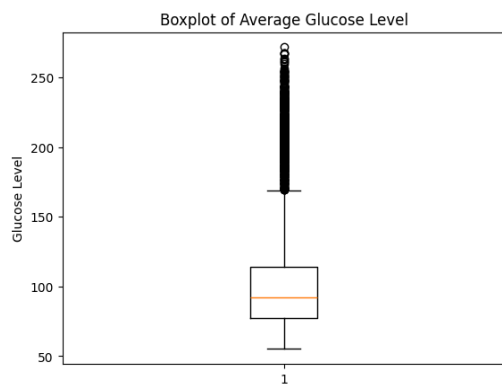


Fig. 2. Average Glucose Level

3. BMI

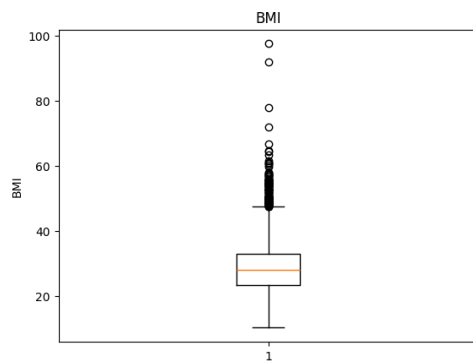


Fig. 3. BMI

C. Statistical Models

1. Age vs Average Glucose Level



Fig. 1. Ave vs Average Glucose Level

2. Average Glucose Level vs BMI

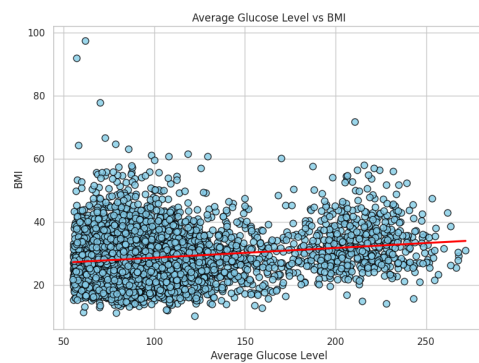


Fig. 2. Age vs Average Glucose Level\

3. Age vs BMI

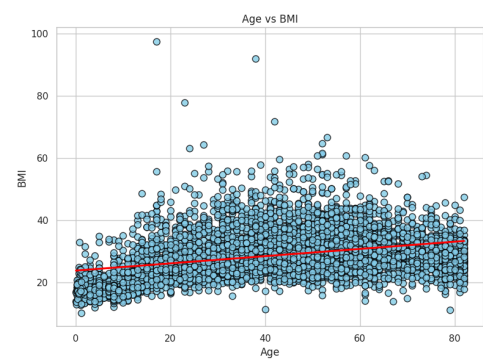


Fig. 3. Age vs BMI

D. Evaluation Metrics

TABLE I MODEL TRAINING

DATA TRAINING				
MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE
DECISION TREE	0.98	0.95	0.47	0.63
RANDOM FOREST	0.95	0.95	1.00	0.97
NEURAL NETWORK HIDDEN LAYER 3 (RE.LU)	0.95	0.95	1.00	0.98
NEURAL NETWORK HIDDEN LAYER 3 (TANH)	0.95	0.95	1.00	0.98
NEURAL NETWORK HIDDEN LAYER 4 (LOGISTIC)	0.95	0.95	1.00	0.98
NEURAL NETWORK HIDDEN LAYER 5 (IDENTITY)	0.95	0.95	1.00	0.98
MULTIPLE HIDDEN LAYER NEURAL NETWORK 4,5	0.95	0.95	1.00	0.98

(RELU)				
MULTIPLE HIDDEN LAYER NEURAL NETWORK 4,5 (TANH)	0.95	0.95	1.00	0.98
MULTIPLE HIDDEN LAYER NEURAL NETWORK 4,4 (IDENTITY)	0.95	0.95	1.00	0.98
MULTIPLE HIDDEN LAYER NEURAL NETWORK 4,5 (LOGISTIC)	0.95	0.95	1.00	0.98

Fig. 1. Data Training Table comparison.

V CONCLUSION

This study demonstrates the potential of machine learning techniques to assist in stroke risk prediction using publicly available health data. By applying classification algorithms such as Linear Regression, Decision Tree, Random Forest, and Neural Network, the study identified key health-related factors including age, hypertension, heart disease, average glucose level, BMI, and smoking status that significantly correlate with the occurrence of stroke. Among the models tested, Neural Network yielded the highest prediction accuracy and robustness, surpassing other methods and highlighting its effectiveness in capturing complex patterns within the data.

Furthermore, descriptive and inferential statistical analyses confirmed the relevance of several independent variables to stroke incidence. The results reinforce the importance of integrating data-driven methods in early detection and decision-making processes in healthcare settings. With proper implementation, these approaches can serve as supporting tools for medical professionals to identify high-risk individuals and intervene early.

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